

Atomic Mean Connectivity Overlay Specification

Plot-D5 averaged atomic net with density overlays

mdstats

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Purpose and status

This document is the normative specification for Plot-D5, introduced in `mdstats` 0.19.35a0 and corrected in 0.19.37a0. Plot-D5 extends the existing framework-dynamics scene with an **averaged atomic connectivity net** rendered in the same 3-D display cell as the mean framework, atomic trajectories, atomic density fields, and framework density fields.

The goal is diagnostic and structural:

1. show the mean atomic bonded geometry in the same registered coordinates used by the density fields;
2. color atomic nodes by chemical species;
3. retain periodic bond connectivity through canonical image shifts; and
4. allow users to toggle species nodes, atomic bonds, and density channels independently in the interactive legend.

Scientific model

Registered atomic positions

Let H_t be the instantaneous cell of frame t and H_d the scene display cell. Atomic vertex positions use the **same registered atomic coordinates as the density fields**. They are not taken from a connectivity-tree lift, because a time-varying spanning tree can change an atom's equivalent periodic image without changing the physical configuration.

For raw fractional atomic coordinate $\mathbf{f}_i(t)$, define

$$\tilde{\mathbf{f}}_i(t) = \begin{cases} \mathbf{f}_i(t), & \text{material coordinates,} \\ \mathbf{f}_i(t) - \mathbf{d}(t), & \text{framework-registered coordinates,} \\ \mathbf{f}_i(t)H_tH_d^{-1}, & \text{laboratory coordinates,} \end{cases}$$

where $\mathbf{d}(t)$ is the mean-framework translational drift used by the framework-dynamics scene.

Because the display cell is periodic and may be triclinic, an ordinary Cartesian average of arbitrarily chosen lattice images is not gauge invariant. The atomic vertex is therefore the weighted periodic Fréchet mean

$$\bar{\mathbf{x}}_i = \arg \min_{\mathbf{x} \in \mathbb{R}^3/\Lambda} \sum_t w_t d_M(\mathbf{x}, \tilde{\mathbf{f}}_i(t) H_d)^2, \quad \sum_t w_t = 1,$$

where d_{MIC} is the Euclidean minimum-image distance induced by H_d . The implemented iteration averages the minimum-image Cartesian displacement vectors and updates the point until convergence. This is a flat-torus specialization of the Fréchet/Karcher center-of-mass construction [1,2].

This definition has two required properties:

1. translating any frame coordinate by an integer lattice vector leaves the result unchanged; and
2. the atomic vertex is centered on the same periodic sample distribution used by the corresponding atomic density field.

Edge occupancy

For an unordered atomic pair $e = (i, j)$ and normalized frame weights w_t , define the bond occupancy

$$p_e = \sum_t w_t \mathbf{1}_{e \in E_t}.$$

Two preparation modes are supported.

Persistent mode

Keep bond e iff it is present in every selected frame,

$$p_e = 1.$$

Occupancy mode

Keep bond e iff

$$p_e \geq p_{\min},$$

where $p_{\min} \in [0, 1]$ is a user-supplied occupancy threshold.

The rendered bond hover text reports the retained occupancy value. The displayed periodic image shift is recomputed from the corrected mean vertex positions by the Euclidean minimum-image convention. Thus connectivity gauge changes cannot create a long or runaway displayed bond.

Public API

New preparation options

```
AtomicMeanGraphOptions(
    mode="occupancy",           # or "persistent"
    occupancy_threshold=0.95,
)
```

New prepared scene payload

```
AtomicMeanGraph(
    atom_indices=...,
    atomic_numbers=...,
    display_positions=...,
    edge_endpoints=...,
    edge_image_shifts=...,
    edge_occupancies=...,
    display_cell=...,
    pbc=...,
    mode=...,
)
```

This object is attached to `FrameworkDynamicsScene.atomic_mean_graph`.

New rendering options

```
AtomicMeanGraph3DRenderOptions(  
    node_size=5.5,  
    node_opacity=0.95,  
    edge_width=2.2,  
    edge_opacity=0.55,  
    edge_color="rgb(120, 120, 120)",  
    show_legend=True,  
)
```

Extended scene-preparation entry point

```
prepare_framework_dynamics_scene(  
    collection,  
    topology,  
    *,  
    atomic_connectivity=...,          # AtomicConnectivityState or Result  
    atomic_mean_graph_options=...,    # AtomicMeanGraphOptions or None  
    ...  
)
```

If `atomic_mean_graph_options` is provided, then `atomic_connectivity` is required.

Extended renderer

```
plot_framework_dynamics_3d(  
    scene,  
    *,  
    atomic_mean_graph_options=AtomicMeanGraph3DRenderOptions(),  
    ...  
)
```

Input constraints

1. The selected scene frames must be covered by the supplied atomic connectivity object.
2. The current implementation requires a **consistent active atom scope** and **consistent atomic numbers** across the selected frames.
3. The current implementation requires a **consistent PBC mask** across the selected frames.
4. The scene display cell must be nonsingular.
5. The connectivity object may be either a single `AtomicConnectivityState` (uniform graph) or an `AtomicConnectivityResult` (time-varying graph).

Rendering policy

Species nodes

One Plotly trace is emitted per atomic species. Species colors are taken from ASE's Jmol-style element color table. Each species trace is legend-toggleable.

Bonds

All retained bonds are emitted in one line trace with `None` separators. Bonds keep their periodic image shift, so cross-boundary bonds remain visible as bonds connecting into neighboring images of the display cell.

Legend behavior

The composite scene layout sets

```
legend={"groupclick": "togglegroup"}
```

so grouped overlays remain easy to toggle in the browser.

Edge cases and limitations

1. The current implementation does not yet support changing active scope across frames, such as atom insertion or removal.
2. The averaged atomic net is most meaningful for relatively persistent bonded structures. Highly reactive trajectories may be better visualized with a more restrictive occupancy threshold.
3. For a strongly multimodal mobile-ion distribution, a single Fréchet mean is still only one summary point and need not represent every occupied site. Use the density cloud and trajectories for transport interpretation.
4. Species toggles and density toggles are independent by design.

Focused validation

The focused tests for Plot-D5 verify the following.

1. occupancy thresholds remove or retain transient edges as expected;
2. a single uniform AtomicConnectivityState is accepted as input;
3. the renderer emits separate species traces plus one bond trace; and
4. Plot-D5 remains compatible with the Plot-D3 and Plot-D4 density tests;
5. connectivity-tree changes cannot move an unchanged atom to another periodic image before averaging; and
6. real Na-LTA validation places all 168 atomic vertices within 0.0011 Å of their periodic registered trajectory means.

References

1. M. Fréchet, “Les éléments aléatoires de nature quelconque dans un espace distancié,” *Annales de l’Institut Henri Poincaré* **10** (1948), 215-310.
2. H. Karcher, “Riemannian center of mass and mollifier smoothing,” *Communications on Pure and Applied Mathematics* **30** (1977), 509-541. DOI: 10.1002/cpa.3160300502.